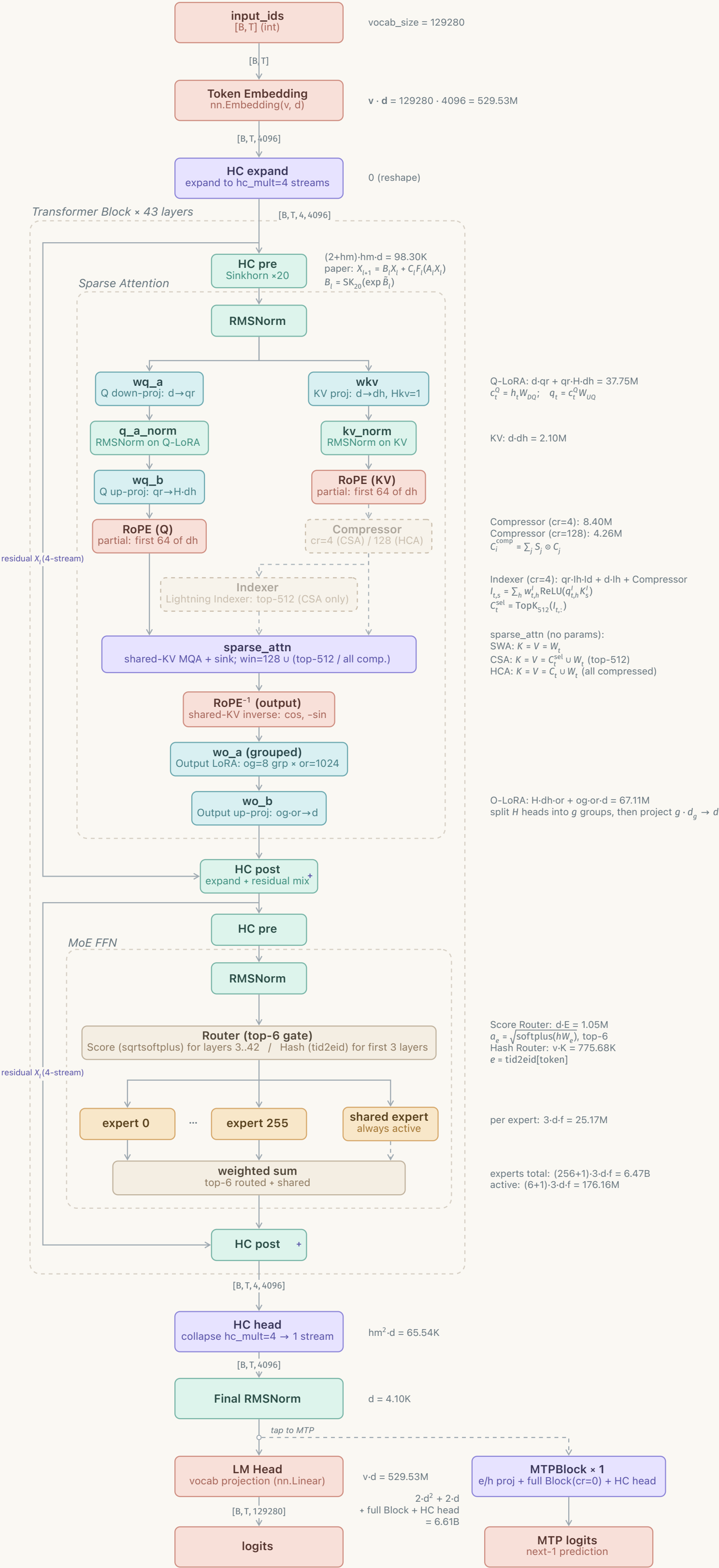


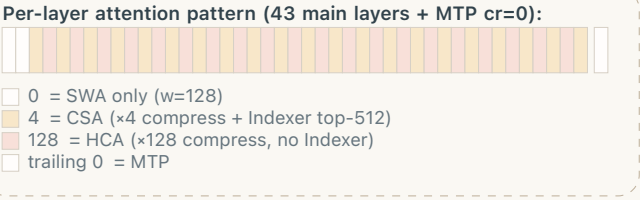
deepseek-ai/DeepSeek-V4-Flash

DSV4 family · open-weight MoE LLM · 1M context · diagram by @yzlnew



Model	deepseek-ai/DeepSeek-V4-Flash
Family	dsv4 (deepseek_v4)
Total params	284.44B (main stack)
Active per token	13.90B
MTP head	6.61B
Total + MTP	291.05B
Quantization	FP8 e4m3, experts=fp4, BF16 gates/norms
Layers (L)	43
Hidden (d)	4096
Heads (H/Hkv)	64 / 1
Head dim (dh)	512
Q LoRA rank	1024
O LoRA / groups	1024 / 8
Routed experts	256 (top-6)
Shared experts	1
MoE FFN dim (f)	2048
HC mult (hm)	4
Index top-K	512
Sliding window	128
Features	hash_routing · moe · mtp shared_experts · sliding_window · swiglu_limit

Symbol legend (attention)		
d	hidden_size	4096
qr	q_lora_rank	1024
H	num_attention_heads	64
dh	head_dim	512
Hkv	num_key_value_heads	1 (shared-KV MQA)
og	o_groups	8
or	o_lora_rank	1024
cr	compress_ratio	0 / 4 / 128 (per-layer)
lh	index_n_heads	64
ld	index_head_dim	128
win	sliding_window	128
topk	index_topk	512 (CSA only)
layer types via compress_ratios[i]:		
0	SWA	sliding window only
4	CSA	+ Compressor + Indexer top-512
128	HCA	+ Compressor (no Indexer)



HC post — shape walk-through (mHC eq. 1)

new_hs

```
= post.unsqueeze(-1) * attn_output.unsqueeze(-2)
+ comb @ hidden_states
```

inputs (4-stream residual X_l):

hidden_states	[B, T, 4, 4096]	X_l
post	[B, T, 4]	C_l
comb	[B, T, 4, 4]	B_l (Sinkhorn)
attn_output	[B, T, 4096]	$F_l(A_l X_l)$

left = $C_l F_l(A_l X_l)$ (sublayer scatter):

```
post.unsqueeze(-1) [B, T, 4, 1]
attn_output.unsqueeze(-2) [B, T, 1, 4096]
(broadcast *) [B, T, 4, 4096]      outer product
```

right = $B_l X_l$ (4×4 stream mix):

```
comb @ hs [B, T, 4, 4096]
```

sum: $X_{l+1} = \text{left} + \text{right}$

```
[B, T, 4, 4096]
```